Modules

Data - 4-11-2026.

Integer Programming.

Math Check #5

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Part 1 - Imagine you have two stocks, A and B.
How much money goes into each?

1. Decision Variable -
For stock A & B, X_A & X_B is the proportion of money invested in stock

Eg - $X_A = 0.25$
= 25% of money goes into stock A

2. Objective function
Max. expected return = $\bar{x}_A X_A + \bar{x}_B X_B$
where →
$\bar{x}_A$ = Monthly Return from stock A
$\bar{x}_B$ = Monthly avg. return from stock B

Eg
$\bar{x}_A = 0.012$ means stock A gained 1.2% in an average month.

3. Constraint: Invest all the money.
i.e. $X_A + X_B = 1$.

Keep risk under the limit

Weighted Covariance matrix:

	A	B
A	$X_A X_B \text{Cov}(A, A)$	$X_A X_B \text{Cov}(A, B)$
B	$X_B X_A \text{Cov}(B, A)$	$X_B X_B \text{Cov}(B, B)$

$$x_A x_A \text{Cov}(A,A) + x_A x_B \text{Cov}(A,B) + x_B x_A \text{Cov}(B,A) + x_B x_B \text{Cov}(B,B) \leq \text{risk limit.}$$

Stay between 0 & 1.

$$0 \leq x_A \leq 1, \quad 0 \leq x_B \leq 1.$$

Part 2 Now add business rules, with many stock: A, B & C and so on. Write each rule & stock A.

6. The new binary decision variable for stock A

New decision variable

$y_A = 1$ (If we buy stock A)

$y_A = 0$ (If we do not)

7. Linking constraint: If we buy stocA, at least 2%

$$x_A \geq 0.02 y_A$$

More Exp.

8. Never more than 10% in stock A

$$x_A \leq 0.10 y_A$$

9. A, B and C are our three Tech stocks. Pick atleat 2 of them.

Equation willbe $\Rightarrow y_A + y_B + y_C \geq 2$

This is a project-selection constraint. It work for any sector and any number: Change the stocks in the sum and the 2.

10. With 10% cap, what is the fewest stocks you can hold and why?

- With a 10% cap you must hold at least 10 stocks (10 x 10% = 100%).